

LOW STYRENE SBR: PATENT LANDSCAPE AND SYNTHESIS ANALYSIS

1. ABSTRACT

This report analyzes patent evidence relevant to low styrene styrene-butadiene rubber (SBR), with emphasis on synthesis routes that control styrene incorporation, polymer microstructure, and downstream processability. The single primary patent in the manifest, US12037432B2, is not a dedicated low-styrene SBR disclosure; rather, it describes a solution-polymerized modified styrene-butadiene copolymer and a post-polymerization stabilization strategy intended to suppress Mooney viscosity increase during steam hot-water treatment and storage. The patent is relevant because it uses anionic solution polymerization of styrene and 1,3-butadiene in hydrocarbon solvent, followed by terminal modification with an alkoxysilane-based modifier and condensation with a modification stabilizer. This approach is representative of modern SBR manufacturing where monomer feed, polar additive selection, initiator choice, and terminal functionalization are used to tune rubber properties.

The disclosed example uses a 20 L autoclave reactor charged with 270 g styrene, 710 g 1,3-butadiene, and 5,000 g n-hexane, with 1.3 g polar additive and 4 mmol n-butyllithium at 40°C. After about 20 minutes, 20 g additional 1,3-butadiene is injected for terminal capping, then 4 mmol of a silane modifier is added, followed by 15 minutes reaction and quench with ethanol and BHT. A subsequent 15-minute condensation with a modification stabilizer is performed before hot-water stripping at 100°C and roll drying. Comparative data show that the stabilizer improves processability metrics, especially Mooney viscosity, while maintaining similar styrene and vinyl contents.

Overall, the evidence supports solution-polymerized SBR as the dominant route for controlled styrene incorporation, but the specific patent does not demonstrate a low-styrene grade; the styrene content reported is about 27 wt%, which is above typical low-styrene targets. The main process trend is post-polymerization stabilization of modified SBR to preserve handling properties after functionalization and thermal workup.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US12037432B2

- Patent Title: Method for preparing modified and conjugated diene-based polymer and modified ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and 1,3-butadiene in a hydrocarbon solvent using an organometallic initiator, followed by terminal modification with an alkoxy silane-based modifier and a post-modification condensation reaction with a modification stabilizer. The polymer is then quenched, stripped in hot water at 100°C, and roll dried to obtain a modified styrene-butadiene copolymer with improved processability. - Reactor volume: 20 L — Example 1 - Styrene charge: 270 g — Example 1 - 1,3-Butadiene charge: 710 g — Example 1 - n-Hexane charge: 5,000 g — Example 1 - Polar additive: 1.3 g of 2,2-di(2-tetrahydrofuryl)propane — Example 1 - Reaction temperature: 40°C — Example 1 - Initiator: 4 mmol n-butyllithium — Example 1 - Terminal capping monomer: 20 g 1,3-butadiene — Example 1 - Modifier amount: 4 mmol of a compound of Formula 5a — Example 1 - Modifier reaction time: 15 minutes — Example 1 - Quench agent: 0.03 parts by weight ethanol based on monomers — Example 1 - Antioxidant addition: 5 mL hexane solution containing 0.3 wt % BHT; 0.03 parts by weight based on monomers — Example 1 - Condensation reaction time: 15 minutes — Example 1 - Hot-water stripping temperature: 100°C — Example 1 - Mooney viscosity: 67 MV — Experimental Example 1, Example 1 - Styrene content: 27.0 wt% based on total polymer amount — Experimental Example 1, Example 1 - Vinyl content: 42.9 wt% based on total polymer amount — Experimental Example 1, Example 1 - Mn: 12.6×10^4 g/mol — Experimental Example 1, Example 1 - Mw: 13.6×10^4 g/mol — Experimental Example 1, Example 1 - PDI: 1.08 — Experimental Example 1, Example 1 - Silica: 70 parts by weight — Experimental Example 2 - Silane coupling agent: 11.02 parts by weight of bis(3-triethoxysilylpropyl)tetrasulfide — Experimental Example 2 - Process oil (TDAE): 33.75 parts by weight — Experimental Example 2 - Antiaging agent (TMDQ): 2.0 parts by weight — Experimental Example 2 - Antioxidant: 2.0 parts by weight — Experimental Example 2 - Zinc oxide: 3.0 parts by weight — Experimental Example 2 - Stearic acid: 2.0 parts by weight — Experimental Example 2 - Wax: 1.0 part by weight — Experimental Example 2 - Banbury mixer speed: 80 rpm — Experimental Example 2 - First-stage discharge temperature: 140°C to 150°C — Experimental Example 2 - Rubber accelerator (CZ): 1.75 parts by weight — Experimental Example 2 - Sulfur powder: 1.5 parts by weight — Experimental Example 2 - Vulcanization accelerator: 2.0 parts by weight — Experimental Example 2 - Second-stage mixing temperature: 60°C or less — Experimental Example 2 - Vulcanization temperature: 180°C — Experimental Example 2 - Vulcanization time: t90+10 minutes — Experimental Example 2

Relevant Experimental Evidence - Example 1: Demonstrated solution polymerization of styrene and butadiene with n-butyllithium, terminal capping, silane modification, and stabilizer

condensation to produce modified SBR. - Example 2: Showed that changing the modification stabilizer to Formula 1b still yields a modified styrene-butadiene copolymer by the same route. - Example 3: Showed that Formula 1c can also be used as the modification stabilizer in the same synthesis sequence. - Comparative Example 1: Demonstrated that omitting the stabilizer condensation step gives a modified SBR with higher Mooney viscosity and poorer processability. - Experimental Example 1: Showed that Examples 1 to 3 retained similar styrene/vinyl contents but had lower Mooney viscosity and lower molecular weights than the comparative sample.

Technical Relevance This patent is relevant because it is a solution-polymerized styrene-butadiene copolymer process with explicit control over styrene incorporation, terminal modification, and post-treatment stability. Although the reported styrene level is not low-styrene, the process principles are directly applicable to designing lower-styrene SBR grades with controlled microstructure and improved handling properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

US12037432B2 shows a solution-polymerized modified SBR route using n-butyllithium, hydrocarbon solvent, polar additive, terminal capping, silane modification, and stabilizer condensation. The evidence indicates that the key synthesis lever is not low styrene content itself, but controlled copolymerization and post-modification stabilization to preserve processability after hot-water stripping. Based on the disclosed data, the most defensible synthesis recommendation for low-styrene SBR development is to retain solution polymerization with precise styrene/butadiene feed control and terminal functionalization, while reducing styrene feed below the reported 27 wt% level and maintaining the stabilizer strategy to limit Mooney viscosity rise during finishing and storage.

5. REFERENCES

- US12037432B2 | Method for preparing modified and conjugated diene-based polymer and modified ... | Not disclosed | US | 2024 | <https://patents.google.com/patent/US12037432B2/en>